RePlastix Innovations: Transforming Plastic Waste into Sustainable Solutions

User Story:

In the RePlastix Innovations system, different users interact with the platform to ensure efficient plastic waste management and recycling. A waste collector uses the system to upload details of collected plastic waste and identify nearby collection centers, so that the waste can be transported and processed efficiently. A citizen or public user interacts with the system to schedule plastic waste pickup and receive rewards or incentives, which encourages regular participation in recycling activities. The user also accesses guidelines on waste segregation to ensure proper disposal.

The recycling plant operator uses the system to monitor the quantity of incoming plastic waste and track the operational status of machines, enabling smooth processing. The operator is also responsible for converting plastic waste into useful products such as tiles, bricks, or fuel, ensuring that waste is effectively reused. The admin manages the overall system by controlling user access, maintaining security, and generating reports on waste collection and recycling performance, which helps in decision-making.

Industrial buyers or companies interact with the system to view and purchase recycled products, supporting eco-friendly practices. Government authorities use the system to track data related to plastic waste reduction and analyze environmental impact, which helps in policy-making and sustainability planning. Additionally, logistics and transport personnel use the system to optimize routes for waste collection and track vehicle movement, improving operational efficiency.

Overall, these user interactions ensure that the RePlastix Innovations system functions as an integrated platform that connects waste collection, recycling, and product utilization, contributing to a sustainable and environmentally responsible solution

1.Introduction:

1.1 Project Overview

Transforming Plastic Waste into Sustainable Solutions

This initiative focuses on shifting from a "take-make-waste" model to a circular economy, where plastic is treated as a valuable resource rather than refuse.

Salesforce

Salesforce +3

Technology-Driven Traceability: Using Salesforce Sales Cloud and Marketing Cloud, organizations can create a fully traceable supply chain. This allows for the verification of the exact amount of plastic prevented from entering oceans, providing the transparency required for Extended Producer Responsibility (EPR) compliance.

Social Impact & Poverty Alleviation: The project model often integrates social benefits. For example, collectors in developing regions can exchange plastic waste for "Social Plastic®" credits, which are redeemable for life-improving essentials like health insurance, school supplies, and digital connectivity.

Operational Waste Reduction: Internally, Salesforce aims for a Nature Positive future by eliminating single-use plastics across its global offices and major events like Dreamforce. This includes using 100% compostable lunch packaging and implementing "Green Angel" volunteer programs to manage waste sorting.

Data-Backed Advocacy: Through partnerships like the Benioff Ocean Initiative, Salesforce uses Tableau to visualize plastic capture data from rivers worldwide. This data is used to influence global policy and infrastructure changes.

Net Zero Cloud Integration: Organizations use Salesforce Net Zero Cloud to track their "waste footprint" alongside carbon emissions. This provides a "single source of truth" for environmental, social, and governance (ESG) reporting.

1.2 Objectives

Establish Full Supply Chain Traceability: Implement a "single source of truth" using Salesforce Sales Cloud to track plastic from the point of collection to its transformation into recycled feedstock. This ensures every kilogram of plastic is verified and auditable.

Scale Social Impact and Poverty Alleviation: Use CRM tools to manage and grow a global marketplace where marginalized community members can exchange collected plastic for secure income and life-improving benefits, such as health insurance and school supplies.

Automate Plastic Offset Management: Streamline the management of plastic credits and offset programs by integrating Salesforce with blockchain platforms. This allows for real-time tracking of partner commitments against actual collection data.

Drive Circular Economy Participation: Facilitate the sale of "Social Plastic®"—recycled plastic feedstock—to corporate manufacturers, helping them replace virgin plastic in their products and meet Extended Producer Responsibility (EPR) requirements.

Optimize Operational Waste Footprints: Use Salesforce Net Zero Cloud to calculate waste-based emissions (Scope 3) and set science-based targets for waste diversion from landfills.

Enhance Data-Driven Decision Making: Deploy Tableau and CRM Analytics dashboards to provide stakeholders with real-time visibility into collection rates, partner engagement levels, and overall environmental impact.

Foster Partner Engagement and Growth: Utilize Marketing Cloud to build personalized onboarding and renewal journeys for corporate partners, increasing the total funds available for plastic collection

1.3 purpose

The purpose of RePlastix Innovations is to:

- Reduce plastic pollution by diverting waste from landfills and oceans.

- Promote sustainability through the creation of reusable, recyclable, and biodegradable alternatives.

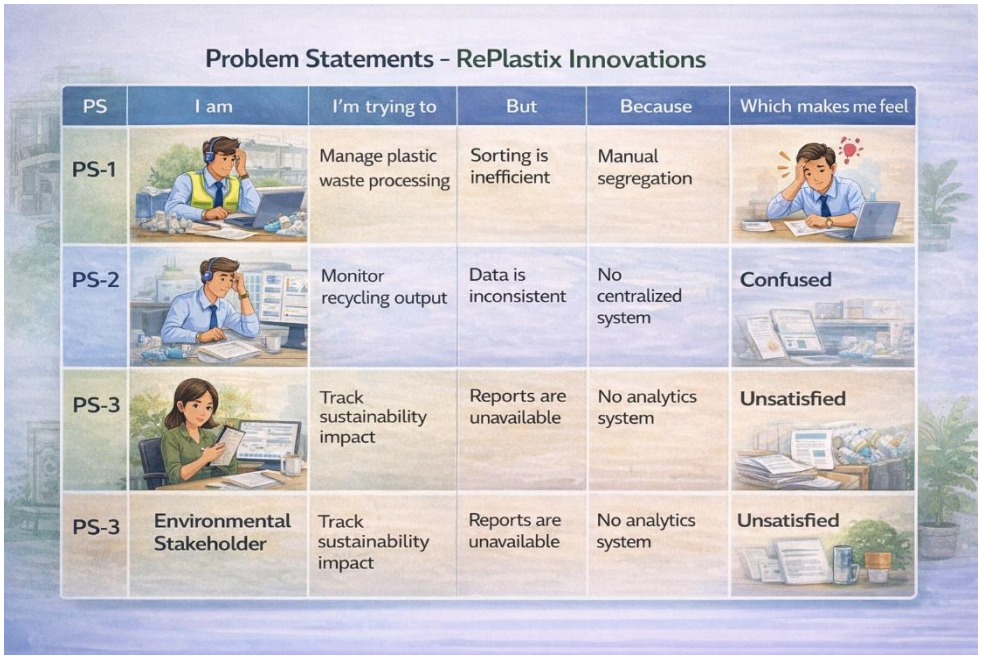
- Encourage community involvement by engaging households, industries, and local organizations in responsible waste disposal and recycling practices.

- Support innovation and employment by developing new eco-friendly products and generating opportunities in the green economy.

2.Ideation phase:

2.1 Problem Statement

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	Recycling Manager	Manage plastic waste processing	Sorting is inefficient	Manual segregation	Frustrated
PS-2	Operations Head	Monitor recycling output	Data is inconsistent	No centralized system	Confused
PS-3	Environmental Stakeholder	Track sustainability impact	Reports are unavailable	No analytics system	Unsatisfied



Empathy Map Canvas

Target Users:

- Recycling Plant Operators
- Environmental Analysts
- Managers
- Stakeholders

Empathy Analysis:

Aspect	Description
Think & Feel	Need efficient waste processing and accurate reporting
Hear	Feedback from environmental agencies
See	Large volumes of unmanaged plastic waste
Say & Do	Manual tracking and reporting
Pain	Inefficiency, pollution, lack of data
Gain	Automation, sustainability tracking

Phase 1: Requirement Analysis & Planning

Understanding Business Requirements:

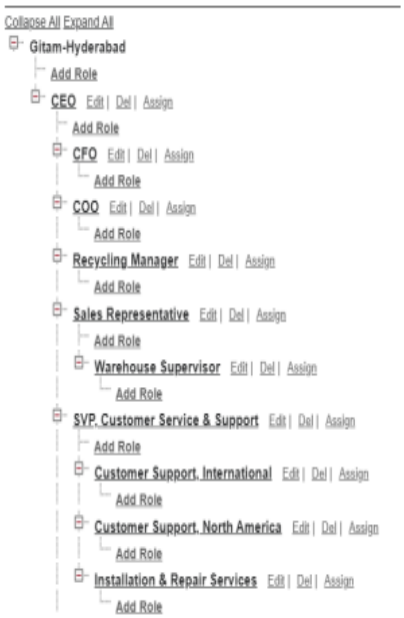
The primary business need was to develop a centralized CRM system for RePlastix Innovations to manage plastic waste recycling operations. Key user requirements included tracking collected plastic waste, monitoring inventory levels of recycled products, automating stock restock processes, and managing customer orders. The system also needed to support departmental access controls, real-time stock notifications, and seamless task assignments based on workflow triggers.

Defining Project Scope and Objectives:

- Create five custom objects: Plastic Waste, Recycling Center, Recycled Product, Order, and Restock Request.
- Enable users to track inventory and automatically generate restock tasks or requests when stock levels fall below thresholds.
- Automate stock updates when restock requests are approved.
- Assign tasks and send email notifications based on workflow conditions.
- Implement validation rules to prevent incorrect or incomplete data entries.
- Ensure role-based data access to support secure information flow across departments.
- Cover all logic with test classes and achieve high code coverage.

Design Data Model and Security Model:

- Data Model: Designed relationships between custom objects using lookups, and defined essential fields like weight, type, collection date, threshold, and stock level.
- Security Model: Configured roles such as CEO, Recycling Manager, Sales Representative, and Warehouse Supervisor. Created custom profiles (Platform 1, 2, 3) with object-level and field-level access based on responsibilities.
- Sharing Rules: Implemented record-level sharing to ensure users only see data relevant to their roles while maintaining overall data security.



Phase 2: Salesforce Development - Backend & Configurations

Setup Environment & DevOps Workflow:

The project was developed using a Salesforce Developer Edition org. I set up the environment by creating a Lightning App named Re Plastix Innovations and configured navigation items for each custom object. All development was done within the Salesforce platform, and version control and test class coverage were maintained using the Developer Console. I also followed a modular approach to testing and debugging Apex logic to ensure scalability and maintainability

Customization of Objects, Fields, Validation Rules, and Automation:

Five custom objects were created to support the business process:

- **Plastic Waste,**
- **Recycling Center,**
- **Recycled Product,**
- **Order, and**
- **Restock Request**

Each object was customized with relevant fields like Weight__c, Threshold__c, Stock_Level__c, Status__c, and Collection_Date__c

Validation rules were implemented to prevent future dates in waste collection and to ensure that order quantity is greater than zero.

Automation was implemented using Flow Builder and Validation Rules: A Scheduled Flow runs daily to check if the stock is below the threshold and creates a highpriority task for the record owner to restock.

- Email notifications are sent to warehouse managers upon restock approval.

No Process Builder or Workflow Rules were used, as Flow Builder is the preferred declarative automation tool in modern Salesforce development.

Apex Classes, Triggers, and Asynchronous Apex:

I developed the InventoryManager Apex class which handles two major operations:

1. **processOrderStock** – Automatically reduces stock levels after an order is placed. If stock is insufficient, it creates a restock request.

2**. processRestockApproval** – When a restock request is approved, the stock is updated accordingly.

Two triggers were implemented:

- UpdateStockAfterOrder – Executes after an order is inserted.
- UpdateStockAfterRestockApproval – Executes after restock status is changed to "Approved".

I also created an EmailNotificationHelper Apex class to send email alerts. Asynchronous Apex was not required in this project as operations were managed synchronously within transaction limits

Phase 3: UI/UX Development & Customization

Lightning App Setup through App Manager:

To provide a unified user experience, I created a custom Lightning App called Re Plastix Innovations using the App Manager. This app includes navigation tabs for the five custom objects—Plastic Waste, Recycling Center, Recycled Product, Orders, and Restock Requests—making it easier for users to access relevant data based on their roles.

Page Layouts and Dynamic Forms:

Each object has been customized with clean and intuitive page layouts, displaying only the most relevant fields to enhance usability. Dynamic forms were configured to show or hide fields based on field values (e.g., hiding restock details unless the stock is low), providing a responsive and context-sensitive user interface.

User Management:

Multiple users were created with assigned roles and custom profiles:

- Recycling Manager (Platform 1)
- Sales Representative (Platform 2)
- Warehouse Supervisor (Platform 3)

Access to records and fields was controlled using profiles, role hierarchy, and sharing rules, ensuring users can only interact with data relevant to their function.

Reports and Dashboards:

Custom reports were created to track:

- Plastic waste collected by type and location
- Stock levels of recycled products
- Pending and approved restock requests
- Order volumes by customer

A dashboard was designed to provide visual insights into KPIs such as stock status, order trends, and restock frequency—supporting data-driven decision-making.

Lightning Pages:

Record pages and home pages were customized using the Lightning App Builder. Components such as record details, related lists, highlights panel, and embedded reports were arranged to deliver a user-friendly and efficient interface

Phase 4: Data Migration, Testing & Security

Data Loading Process:

Initial data for plastic waste, recycling centers, products, orders, and users was imported using the Data Import Wizard for simple object-level imports. For more complex or relational data, such as linking orders to products or assigning lookup relationships, Data Loader was used to ensure accuracy and bulk insert capability.

Field History Tracking, Duplicate Rules, Matching Rules:

- Field History Tracking was enabled for key fields such as Stock_Level__c, Threshold__c, and Status__c in the Recycled Product and Restock Request objects, helping to monitor changes over time for audit and analysis.
- Duplicate Rules and Matching Rules were configured to avoid duplicate entries for products and orders. For example, duplicate recycled product names or identical customer order entries trigger a warning to the user.

Profiles, Roles, Role Hierarchy, Permission Sets, Sharing Rules:

• Roles: CEO (top-level), Recycling Manager, Sales Representative, Warehouse Supervisor

• Profiles: Platform 1 (limited access), Platform 2 (moderate access), Platform 3 (full access)

• Sharing Rules: Set up for record-level access between these roles. For example: o CEO → Sales Rep (Read Recycled Products) o Sales Rep → Warehouse Supervisor (Read Restock Requests)

• Permission Sets were used to grant additional access temporarily for testing users beyond profile-level access, ensuring flexibility in permission management.

Creation of Test Classes:

• Apex class InventoryManagerTest was written to test the business logic defined in InventoryManager.

• It validated:

o Stock deduction after order

o Restock request creation

o Stock addition after approval

• The test class achieved 100% code coverage, confirming that all scenarios are handled and that the triggers and methods behave as expected

Preparation of Test Cases:

For each Salesforce feature, detailed test cases were prepared with inputs, expected outputs, and screenshots. Here are the highlights:

Feature	Test Case	Expected Output
Booking (Order) Creation	Input a quantity > available stock	Restock Request auto-created; Product stock reduced
Approval Process	Approve Restock Request	Product stock auto-updated
Automatic Task Creation (Flow)	Lower product stock below threshold	Task created for record owner with subject: "Stock Low - Fill ASAP"
Triggers	Create Order / Approve Restock	Triggers fire and update logic executes correctly
Validation Rules	Set quantity = 0 or collection date in future	Error message displayed; record blocked
Sharing Rules	Login as different roles (Sales Rep, Manager, etc.)	Users only see permitted records and fields
Email Notification	Approve a restock request	Warehouse Manager receives notification email

Phase 5: Deployment, Documentation & Maintenance

Deployment Strategy:

For deployment, a manual migration approach was followed since the project was developed in a single Salesforce Developer Org for demonstration purposes. However, in a real-world multi-org environment, deployment would typically be handled using:

- Change Sets (for deploying from Sandbox to Production)
- Salesforce CLI (for larger or automated deployments)
- Version control systems like Git to track Apex classes, flows, triggers, and metadata

. All components including custom objects, fields, Apex classes, validation rules, and flows were packaged and verified for readiness before deployment

System Maintenance & Monitoring:

To ensure the system remains functional and scalable:

- Scheduled Flows and automated tasks are regularly monitored for success/failure from the Flow UI
- Debug Logs and Apex Exception Logs are used for monitoring automation and logic errors
- Field History Tracking supports audit trails on key objects like Recycled Products and Restock Requests
- Periodic reviews of user permissions and sharing rules are conducted to ensure data security and proper access

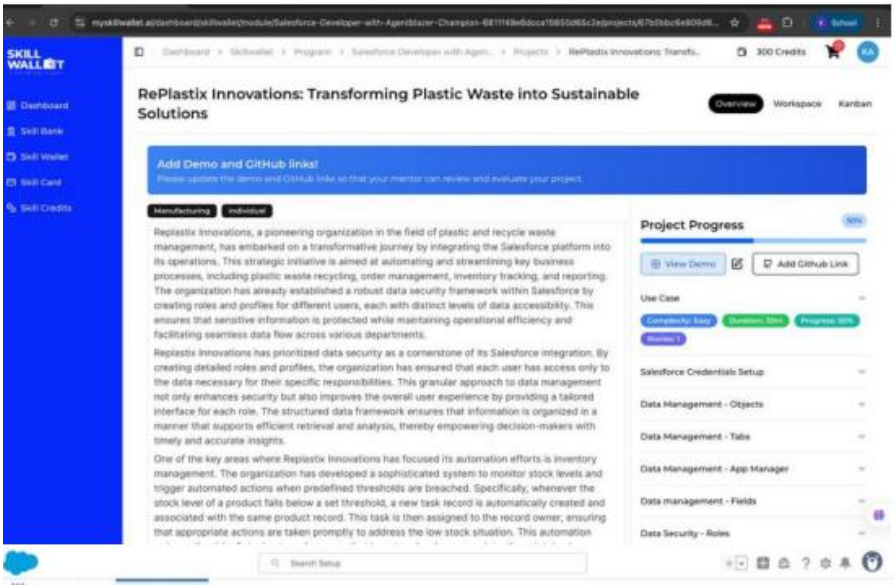
Additionally, regular updates or enhancements like new validation rules, reports, or automated processes can be added with minimal disruption to existing features, using sandbox testing before live implementation.

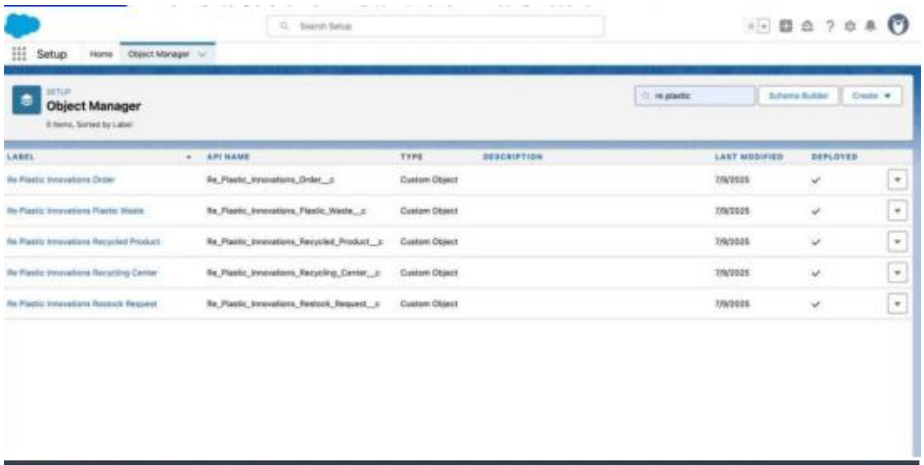
Troubleshooting Approach & Documentation:

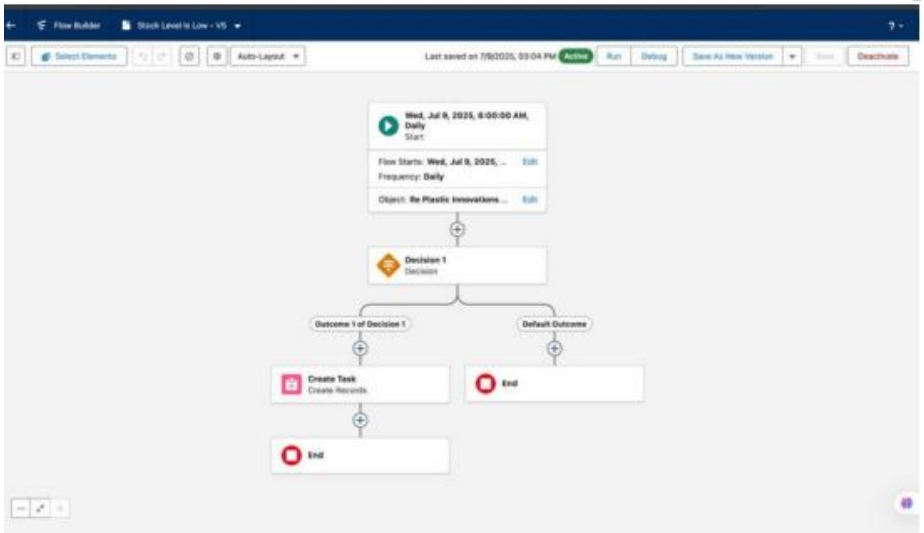
A troubleshooting guide was created to assist with resolving common issues:

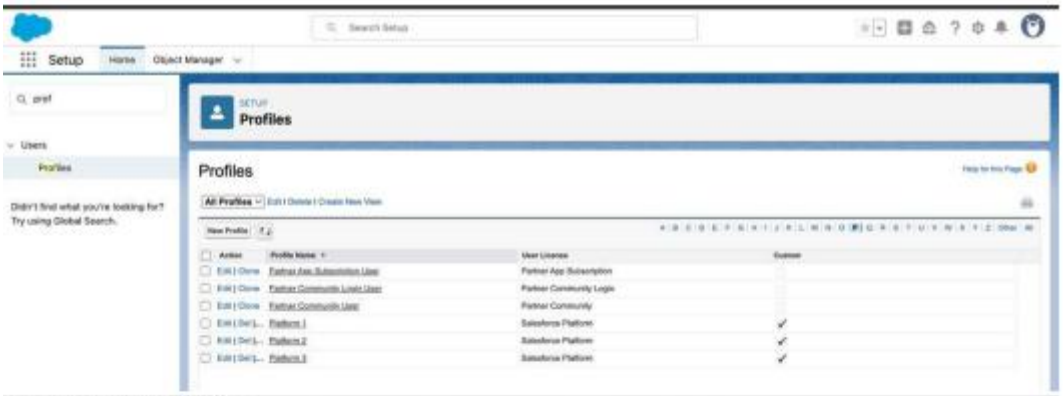
- Error in Apex Logic or Triggers: Use Developer Console logs to trace exception lines
- Validation Errors: Review field-level and object-level validation logic under Object Manager → Validation Rules
- Flow Failures: Use Flow Interview Logs and the Paused Flow Interviews list to identify failed flows or tasks
- Permission Issues: Check profile and permission set access, verify role hierarchy and sharing rules
- Data Loading Errors: If using Data Loader, error files are reviewed for each upload to track failures (e.g., missing lookups, required fields)

This structured documentation ensures future users and admins can maintain the system efficiently and resolve issues quickly without deep developer intervention


SKILL WALLET
Dashboard
Skill Bank
Skill Wallet
Skill Card
Skill Credits
300 Credits
RePlastix Innovations: Transforming Plastic Waste into Sustainable Solutions
Overview
Workspace
Kanban
Add Demo and GitHub links!
Please update the demo and GitHub links so that your mentor can review and evaluate your project.
Manufacturing
Individual
Replastix Innovations, a pioneering organization in the field of plastic and recycle waste management, has embarked on a transformative journey by integrating the Salesforce platform into its operations. This strategic initiative is aimed at automating and streamlining key business processes, including plastic waste recycling, order management, inventory tracking, and reporting. The organization has already established a robust data security framework within Salesforce by creating roles and profiles for different users, each with distinct levels of data accessibility. This ensures that sensitive information is protected while maintaining operational efficiency and facilitating seamless data flow across various departments.
Replastix Innovations has prioritized data security as a cornerstone of its Salesforce integration. By creating detailed roles and profiles, the organization has ensured that each user has access only to the data necessary for their specific responsibilities. This granular approach to data management not only enhances security but also improves the overall user experience by providing a tailored interface for each role. The structured data framework ensures that information is organized in a manner that supports efficient retrieval and analysis, thereby empowering decision-makers with timely and accurate insights.
One of the key areas where Replastix Innovations has focused its automation efforts is inventory management. The organization has developed a sophisticated system to monitor stock levels and trigger automated actions when predefined thresholds are breached. Specifically, whenever the stock level of a product falls below a set threshold, a new task record is automatically created and associated with the same product record. This task is then assigned to the record owner, ensuring that appropriate actions are taken promptly to address the low stock situation. This automation
Project Progress
50%
View Demo
Add Github Link
Use Case
Complexity: Easy
Duration: 30m
Progress: 50%
Stories: 1
Salesforce Credentials Setup
Data Management - Objects
Data Management - Tabs
Data Management - App Manager
Data management - Fields
Data Security - Roles
Search Setup


Search Setup
Setup
Home
Object Manager
SETUP
Object Manager
8 items, Sorted by Label
re plastic
Schema Builder
Create
LABEL
API NAME
TYPE
DESCRIPTION
LAST MODIFIED
DEPLOYED
Re Plastic Innovations Order
Re_Plastic_Innovations_Order__c
Custom Object
7/9/2025
Re Plastic Innovations Plastic Waste
Re_Plastic_Innovations_Plastic_Waste__c
Custom Object
7/9/2025
Re Plastic Innovations Recycled Product
Re_Plastic_Innovations_Recycled_Product__c
Custom Object
7/9/2025
Re Plastic Innovations Recycling Center
Re_Plastic_Innovations_Recycling_Center__c
Custom Object
7/9/2025
Re Plastic Innovations Restock Request
Re_Plastic_Innovations_Restock_Request__c
Custom Object
7/9/2025


Flow Builder
Stock Level is Low - V5
Select Elements
Auto-Layout
Last saved on 7/9/2025, 03:04 PM
Active
Run
Debug
Save As New Version
Save
Deactivate
Wed, Jul 9, 2025, 6:00:00 AM, Daily
Start
Flow Starts: Wed, Jul 9, 2025, ...
Edit
Frequency: Daily
Object: Re Plastic Innovations ...
Edit
Decision 1
Decision
Outcome 1 of Decision 1
Default Outcome
Create Task
Create Records
End
End

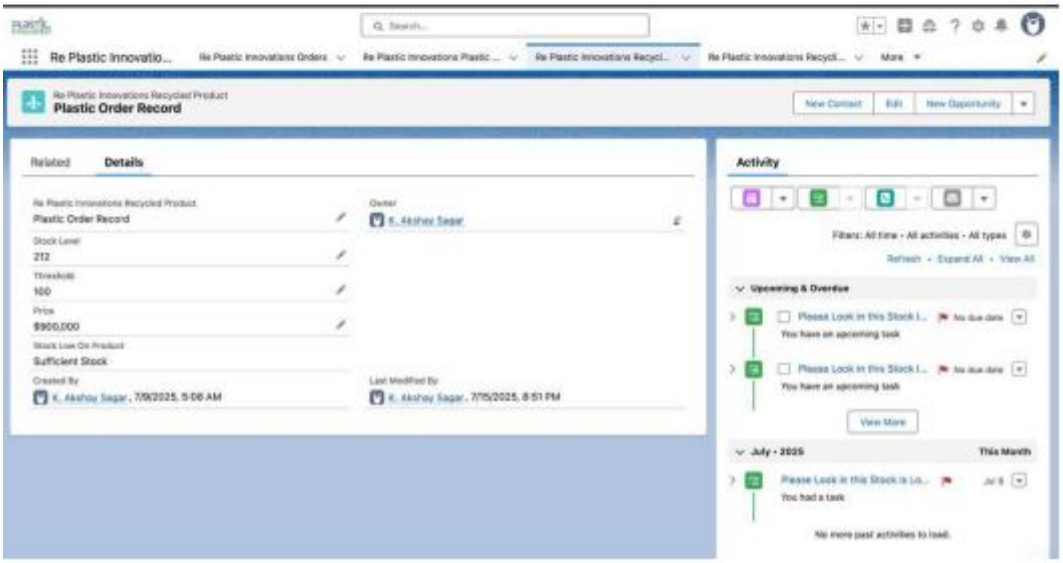



```
@isTest
public class InventoryManagerTest {

    @testSetup
    static void setupTestData() {
        // Create product
        Re_Plastic_Innovations_Recycled_Product__c product = new Re_Plastic_Innovations_Recycled_Product__c(
            Stock_Level__c = 50,
            Threshold__c = 10
        );
        insert product;

        // Create orders
        List<Re_Plastic_Innovations_Order__c> orders = new List<Re_Plastic_Innovations_Order__c>{
            new Re_Plastic_Innovations_Order__c(Recycled_Product__c = product.Id, Quantity__c = 20),
            new Re_Plastic_Innovations_Order__c(Recycled_Product__c = product.Id, Quantity__c = 40)
        };
        insert orders;
    }

    @isTest
    static void testProcessOrderStock() {
```



3. REQUIREMENT ANALYSIS PHASE

Replastix Innovations: Transforming Plastic Waste into Sustainable Solutions

The Requirement Analysis Phase identifies the needs of users and defines system functionalities. For the Replastix Innovations system, this phase focuses on understanding how plastic waste is collected, processed, tracked, and converted into sustainable products using a digital platform.

3.1 Customer Journey Map

The Customer Journey Map describes the steps users follow when interacting with the Replastix platform.

Stage

- Awareness
- Registration
- Plastic Waste Submission
- Waste Collection
- Recycling Process
- Product Purchase
- Feedback

This journey helps ensure a smooth experience for waste contributors, recycling teams, and product buyers.

3.2 Functional Requirements

Functional requirements define what the system should do.

1. User Registration and Login

Users must be able to create accounts and log in securely.

2. Waste Submission Management

Users should be able to schedule plastic waste pickups or report drop-off locations.

3. Waste Tracking

The system should track plastic waste from collection to recycling.

4. Admin Management

Admin can monitor waste collection, recycling progress, and product inventory.

5. Product Marketplace

Customers should be able to purchase recycled products online.

6. Notification System

The system sends notifications for pickup schedules, recycling updates, and orders.

7. Feedback and Reporting

Users can submit feedback and view environmental impact reports.

Non-Functional Requirements

NFR-1: Usability

Requirement:

The system should provide a user-friendly interface that allows users to easily submit plastic waste and track recycling activities.

Example:

A mobile-friendly application where users can schedule a plastic pickup in less than 3 simple steps.

NFR-2: Security

Requirement:

The system must ensure protection of user data, payment information, and transaction details.

Example:

User passwords stored using encryption (hashed passwords).

Secure payment gateway integration.

NFR-3: Reliability

Recuirement:

The system should operate consistently and perform recycling data tracking without failures.

Example:

If a pickup request is submitted, the system must reliably store the request and notify the collection team without data loss.

NFR-4: Availability

Requirement:

The system should be available for users at all times with minimal downtime.

Example:

The platform should maintain 99% uptime so users can schedule pickups or purchase recycled products anytime.

NFR-5: Scalability

Requirement:

The system should be able to handle increasing numbers of users, waste submissions, and product orders.

Example:

If the number of users grows from 1,000 to 50,000, the system should still maintain good performance without slowing down.

3.3 Technology Stack

Layer	Technology
Frontend	Web Interface
Database	Cloud Storage
Backend	Processing System
Automation	Workflow Engine
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Traditional plastic waste management is inefficient, leading to pollution and resource wastage.

The proposed system:

- Automates waste tracking
- Improves recycling efficiency
- Provides real-time insights
- Reduces environmental impact

4.2 Proposed Solution

Functional Components

Parameter	Description
User Login & Authentication	Secure access for users
Waste Data Management	Store and manage plastic waste details
Waste Classification	Categorize plastic types
Recycling Process Tracking	Monitor processing stages
Production Monitoring	Track recycled outputs
Validation Rules	Ensure accurate data
Notification System	Alerts for operations
Workflow Automation	Automates recycling steps
Report Generation	Generate performance reports
Dashboard Visualization	Real-time insights
Data Backup & Recovery	Secure storage
User Role Management	Controlled access

4.3 Solution Architecture

Architecture Layers:

1. Presentation Layer – User Interface
2. Application Layer – Processing Logic
3. Data Layer – Waste & Production Data
4. Analytics Layer – Reports & Dashboards

This ensures scalability, efficiency, and maintainability.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile-based development approach
- System designed for waste tracking and recycling
- Database for plastic waste categories
- Workflow automation for recycling stages
- Dashboard for environmental insights
- Regular testing and monitoring

Product Backlog & Sprint Schedule

Sprint	Functional Requirement	User Story Number	User Story	Story Points	Priority	Team
Sprint-1	Setup	USN-1	Create system environment	1	High	Member1
Sprint-1	Data Modeling	USN-2	Create waste database	3	High	Member2
Sprint-2	Data Modeling	USN-3	Create waste categories	3	High	Member3
Sprint-2	UI Design	USN-4	Develop user interface	3	High	Member4
Sprint-3	Automation	USN-5	Automate waste processing	5	High	Member1
Sprint-3	Validation	USN-6	Validate waste inputs	5	High	Member2

Sprint	Functional Requirement	User Story Number	User Story	Story Points	Priority	Team
Sprint-4	Processing	USN-7	Track recycling stages	5	Medium	Member3
Sprint-5	Reports	USN-8	Generate sustainability reports	4	High	Member4
Sprint-6	Dashboards	USN-9	Create analytics dashboards	4	Medium	Member1

Project Tracker

Sprint	Story Points	Duration	Start Date	End Date	Completion
Sprint-1	20	6 Days	Day 1	Day 6	Completed
Sprint-2	20	6 Days	Day 7	Day 12	Completed
Sprint-3	20	6 Days	Day 13	Day 18	Completed
Sprint-4	20	6 Days	Day 19	Day 24	Completed
Sprint-5	20	6 Days	Day 25	Day 30	Completed

6.Project Development Phase:

Project Flow

Milestone 1 : creation of developer account

Milestone 2 : Object creation

Milestone 3 : Tabs

Milestone 4 : App manager

Milestone 5 : Fields

Milestone 6 : Roles

Milestone 7 : Profiles

Milestone 8 : Create users

Milestone 9 : Record level setting

Milestone 10 : Data configuration

Milestone 11 : Flow builder

Milestone 12 : apex

Implementation

Milestone 1 : creation of developer account

Activity 1 : Developer account creation

Creating a developer org in salesforce.

1.Go to https://developer.salesforce.com/signup

On the sign up form, enter the following details :

1.First name & Last name

2.Email

3.Role : Developer

4.Company : College Name

5.County : India

6.Postal Code : pin code

7.Username : should be a combination of your name and company

This need not be an actual email id, you can give anything in the format : username@organization.com

Click on sign me up after filling these.

Activity 2 : **Account Activation**

1.Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.

2.Click on Verify Account

3.Give a password and answer a security question and click on change password.

4.Then you will redirect to your salesforce setup page.

Milestone 2 : Object creation

Objects:

Re Plastic Innovations Plastic Waste(Re_Plastic_Innovations_Plastic_Waste)

Field API Name	**Data Type**	**Description**
Name	Text (Auto Number)	Unique ID for waste records
Weight__c	Number (18,2)	Weight of plastic waste (kg)
Type__c	Picklist	Type of plastic (PET, HDPE, PVC, etc.)
Collection_Date__c	Date	Date waste was collected

Status__c	Picklist	["Collected", "Processing", "Recycled"]
Recycling_Center__c	Lookup (Recycling_Center__c)	Assigned recycling center
Location__c	Geolocation	Waste collection location

Re Plastic Innovations Recycling Center (Re_Plastic_Innovations_Recycling_Center__c)

Field API Name	Data Type	Description
Name	Text	Recycling Center Name
Location__c	Geolocation	Center’s location
Capacity__c	Number (18,2)	Max capacity for processing waste

Re Plastic Innovations Recycled Product (Re_Plastic_Innovations_Recycled_Product__c)

Field API Name	Data Type	Description
Name	Text	Recycled product name
Stock_Level__c	Number	Current stock available
Threshold__c	Number	Minimum stock before restock is triggered
Price__c	Currency	Price per unit

Re Plastic Innovations Order (Re_Plastic_Innovations_Order__c)

Field API Name	Data Type	Description
Name	Auto Number	Order ID
Customer__c	Lookup (Account)	Customer placing the order
Recycled_Product__c	Lookup (Recycled_Product__c)	Ordered product
Quantity__c	Number	Quantity ordered
Delivery_Date__c	Date	Expected delivery date

Re Plastic Innovations Restock Request (Re_Plastic_Innovations_Restock_Request__c)

Field API Name	Data Type	Description
Name	Auto Number	Request ID
Product__c	Lookup (Recycled_Product__c)	Product to restock
Requested_Quantity__c	Number	Quantity requested
Status__c	Picklist	["Pending", "Approved", "Completed"]

Activity 1 : Object - Re Plastic Innovations Plastic Waste

To Navigate to Setup page:

Click on gear icon → click setup.

The purpose of creating a Re Plastic Innovations Plastic Waste object is to have a clear picture of the Re Plastic Innovations Plastic Waste.

To create an object:

1.From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.

2.Enter the label name→ Re Plastic Innovations Plastic Waste

3.Plural label name→ Re Plastic Innovations Plastic Wastes

4.Enter Record Name Label and Format

Record Name → Plastic Waste Name

Data Type → Auto Number

5.Click on Allow reports,

6.Allow search → Save.

Activity 2 : Object - Re Plastic Innovations Recycling Center

The purpose of creating a Dealer object is to have detailed information about Dealer.

To create an object:

1.From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.

2.Enter the label name→ Re Plastic Innovations Recycling Center

3.Plural label name→ Re Plastic Innovations Recycling Centers

4.Enter Record Name Label and Format

Record Name → Re Plastic Innovations Recycling Center

Data Type → Text

5.Click on Allow reports.

6.Allow search → Save

Activity 3 : create other objects

Creating remaining objects as mentioned above.

Milestone 3 : Tabs

Activity 1 : Tab - Re Plastic Innovations Plastic Waste

Creating a Custom Tab(Re Plastic Innovations Recycling Center)

1.Go to setup page → type Tabs in Quick Find bar → click on tabs → New (under custom object tab)

2.Select Object(Re Plastic Innovations Plastic Waste) → Select any tab style → Next (Add to profiles page) keep it as default → Next (Add to Custom App) keep it as default → Save.

Activity 2 : Create Tabs

Now create tabs for Other objects as well which we have created above Objects

Milestone 4 : App manager

Activity 1 : App Replastic innovation

To create a lightning app page:

1.Go to setup page → search “app manager” in quick find → select “app manager” → click on New lightning App.

Fill the app name in app details and branding as follow

1.App Name : Re Plastic Innovations

2.Developer Name : this will auto populated

3.Description : Give a meaningful description

4.Image : optional (if you want to give any image you can otherwise not mandatory)

5.Primary color hex value : keep this default

6.Then click Next → (App option page) keep it as default → Next → (Utility Items) keep it as default → Next.

To Add Navigation Items:

1.Search the items in the search bar(Restock Requests, Recycling Centers, Recycled Products, Plastic Wastes, Orders) from the search bar and move it using the arrow button → Next.
Note: select the custom object which we have created in the previous activity.

2.To Add User Profiles:

Search profiles (System administrator) in the search bar → click on the arrow button → save & finish.

Milestone 5 : Fields

Activity 1 : Field-weight

To create fields in an object:

1.Go to setup → click on Object Manager → type object name(Re Plastic Innovations Plastic Waste) in quick find bar→ click on the object.

2.Now click on “Fields & Relationships” → New

3.Select Data type as “Number”.

4.Click on Next

Fill the above as following:

1.Field Label: Weight

2.Length : 10

3.Click on Next → Next → Save and new.

Activity 2 : Field-Type

Create Second Filed in Re Plastic Innovations Plastic Waste

1.Select Data type as “PickList”.

2. Click on Next

3. Fill the above as following:

Field Label: Type

Picklist Values : (PET, HDPE, PVC, etc.)

Click on Next → Next → Save and new.

Activity 3 : Field - Re Plastic Innovations Recycling Center

Create Lookup RelationShip in Re plastic Innovations Plastic Waste Object to Steel Production Object.

1.Select DataType as Lookup Relationship.

2.Click Next

3.Select Re Plastic Innovations Recycling Center Object Form Related Filed.

4.Label Name Re Plastic Innovations Recycling Center and Click Next

5.Click Next and Save.

Activity 4 : Create other fields

Creating remaining all the fields for the objects which we have created in Data Management -Object.

Milestone 6 : Roles

Activity 1 : Role-Recycling Manager

Creation Roles

1.From the setup menu, search and select roles

2.Click Add Roles CEO Below Option.

Label : Recycling Manager

This Role Reports To : CEO

Click Save and New

Activity 2 : Creaete other roles

Create More Two Roles

1. Sales Representative report to CEO
2. Warehouse Supervisor report to Sales Representative

Milestone 7 : Profiles

Activity 1 : Profile platform 1

Create Custom Profile

1.From Setup page, search and select profiles --> click on new Profile

2.Fill the details as shown below:

Existing Profile Name : Standard Platform user

User License : Salesforce platform

Profile Name : Platform 1

3.Click Save

Activity 2 : Create other profiles

Create More Two Profiles

1.Platform 2

2.Platform 3

Activity 3 : Modify platform 1

1.Click Edit on the profile 1

2.Add Access From This Platform 1 Profile and Save It.

3.Access Objects

4.Re Plastic Innovations Plastic Waste : Read/Create

5.Re Plastic Innovations Restock Request : Read-Only

Activity 4 : Modify other profiles

Give Two More Profile Object Level Access

1.Profile : Platform 2

Read/Create: Re Plastic Innovations Order , Account

Read-Only: Re Plastic Innovations Plastic Waste, Re Plastic Innovations Recycled Product

2.Profile : Platform 3

Read/Create/Edit : All Custom objects

Milestone 8 : Create users

Activity 1 : User - John Production Engineer

User Creation

1.From the setup page, search and select users --> click New User

2.Fill Out User Information: You'll need to provide the following details for the new user:

- First Name: John Production Engineer
- Last Name: Sandbox 1.
- Alias: Auto Populate
- Email: Give your Personal Email id
- Username: Username should be in this form: text@text.text (JohnProductionEngineer@sandbox1.com).
- Role: Recycling Manager
- Profile: Platform 1
- License: Salesforce Platform

3.Click Save

4.Check Gmail for Reset Password for New user

5.Click Verify Account

6.Click Reset Password

Set your Password

Click Change Password

Activity 2 : Create user

Create Two More users

1.User 2 :

- First name : Quality Inspector
- LastName : Mike
- Email id : Give your Personal Email id
- Username : Username should be in this form: text@text.text
- Role : Sales Representative
- Profile : Platform 2

3.User : 3

- FIrst Name : Plant Manager
- Last Name : Albert
- Email id : Give your Personal Email id
- Username : Username should be in this form: text@text.text
- Role : Warehouse Supervisor
- Profile : Platform 3

Milestone 9 : Record level setting

Activity 1 : Sharing Settings - Recycling Manager Access

Give Record Level Access To Users

1. Search Sharing Setting From Quick Find Box and Click Edit

2.Make OWD (Organizational Wide Default) Access For Re Plastic Innovation objects

3.Click Save

4.Give Record Level Access To User

Only the Recycling Manager should access Re_Plastic_Innovations_Plastic_Waste__c collection details.
Search Steel Production Object and Click New for Crete New Sharing Rule.

- Label Name : Recycling Manager Access
- Rule Name : Recycling_Manager_Access
- Select Which Record Should Be Shared : CEO
- Select the Users to share With : Recycling Manager
- Access Level : Read Only

5.Click Save

Activity 2 : create sharing setting

Create More Two Sharing Rules

1.Sales Representative get read access for Re_Plastic_Innovations_Recycled_Product Object.

CEO ====> Sales Representative.

2.Warehouse Supervisor gets read access for Re_Plastic_Innovations_Restock_Request Object.

Sales Representative.=====> Warehouse Supervisor

Milestone 10 : Data configuration

Activity 1 : Formula Field - Stock Low On Product

1.Select Object From Quick Find Box Object Name Is Re Plastic Innovations Recycled Product

2.Click Fields and Relationships and Click New Button For creation of Formula Field

3.Select DataType Formula Filed and Click Next

Label Name : Stock Low On Product

DataType : Text

4.Click Next

5.Click Advance Formula and Write Formula logic Is Stock less than with Therhold set Value

6.Formula : IF(Stock_Level__c < Threshold__c, "Low Stock - Restock Needed", "Sufficient Stock")

7.Test the Logic Create Record

8.Change Stock Value

Activity 2 : Validation Rule - Check_Quantity_Not_Zero

Create Validation Rule

1.Search Re Plastic Innovations Order Object From Quick Find Box

2.Select Validation Rule and Click New to Create Validation Rule

- Rule Name : Check_Quantity_Not_Zero
- Active : True
- Formula : Quantity__c <=0
- Error Message : "Quantity must be greater than zero."
- Filed : Quantity

3.Click Save

4.Test Record

5.Change Quality to more than 0

6.Create Another Validation on Re Plastic Innovations Plastic Waste

- **Name:** Future_Date_Collection
- Active : True
- Formula : Collection_Date__c > TODAY()
- **Error Message:** "Collection Date cannot be in the future."
- Filed : Collection_Date__c

Milestone 11 : Flow builder

Activity 1 : Flow Builder

1.Go to Setup (Click on the gear icon in the top-right corner → Select Setup).

2.In the Quick Find Box, type Flows and click on Flows and Click New Flow

3.Select Start From Scratch then Click Next

4.Select Schedule Triggered Flow Then Click Create

5.Set Time and Date and This Flow It need run Daily

- Date : Today Date
- Time : 6:00Am
- Frequency : Daily

6.Select Object

7.Select Re_Plastic_Innovations_Recycled_Product__c Object

8.Click Plus Icon For Add Decision Element

- Label Name : Decision 1
- Out Come Details : Outcome 1 of Decision 1
- Conditions : {!$Record.Stock_Level__c} LessThan {!$Record.Threshold__c}

9.Select Puls Icon and Search For Create Record For Task object Element

- Label : Create Task
- How to Set Field Values : Manually
- Create a Record Of the Object : Task
- Set Filed Values for the task:
- Assigned To ID : {!$Record.OwnerId}
- Related To Id : {!$Record.Id}
- Priority : High
- Subject : Please Look in this Stock Is Low Fill The Stock ASAP
- Status : In Progress

10.Click Save Label Name as "Stock Level Is Low" and Click Save.

11.Then Activate Flow

12.Test The Flow Set Time and Create Record and Check It.

13.Click on Upcoming & OverDue.

Milestone 12 : apex

Activity 1 : Create Apex Class

1. Click Gear Icon and Select Developer Console
2. Click File and Click New and Select Apex class
3. Create an Apex Class and name it as "InventoryManager" and Click Ok
4. Write Code Inside Class

Create an Apex Trigger:

1. Call the Apex Class In Triggers Click File and Click New and Select Apex Trigger
2. Write an Apex Trigger and name it as "UpdateStockAfterOrder".
3. Select Re_Plastic_Innovations_Order__c as Object and Click Submit.
4. Inside TriggerClass Write Code

Create another Apex Trigger:

1. when the stock is low in the Re_Plastic_Innovations_Recycled_Product__c object. The trigger should update the stock via the Re_Plastic_Innovations_Order__c object and send the request to the Re_Plastic_Innovations_Restock_Request__c object. If the request is approved, the stock should automatically be updated in the Re_Plastic_Innovations_Recycled_Product__c object
2. Trigger Name is "UpdateStockAfterRestockApproval" and object is Re_Plastic_Innovations_Restock_Request__c click Submit
3. Write Code Insider Trigger Class

4. Send Email To warehouse Manager Stock is Updated Successfully you can Create Stock
5. Create Apex Class In that we can Write Email Logic then we need to call this Apex in Trigger Class
6. Apex Class name is EmailNotificationHelper and Click Ok
7. write Code inside the Apex Class
8. Call this Apex class in Triggers Class(UpdateStockAfterRestockApproval)

Test:

1. How to Test this, First Create Record on Re_Plastic_Innovations_Recycled_Product__c object Set Stock_Level__c = 50 and Threshold__c = 100 and save the Record.
2. Create Re_Plastic_Innovations_Order__c Object Record Select Recycled_Product parent Record(the record which we created above(Plastic Order Record)) and Quatity as 150 and Save the Record.
3. Check Related Re Plastic Innovation Orders Record in Selected Recycled_Product__c's record (Plastic Order Record) Object it will also Creates a Request record for update stock in Selected Recycled_Product__c Record.
4. Now click on the newly created Re Plastic Innovations Restock Request record from the related list page and notice the status is pending .
5. Change the status to approve and Save the Record.
6. Check the Record Select Recycled_Product__c record(Plastic Order Record, which we have created at the begining) stock is updated more numbers.

And User will Get Email

1. Write Test class For Apex CLass the Test class will cover Trigger class As well.
2. Class Name is InventoryManager to Write Test Class
3. Write Test Class Name InventoryManagerTest and Click Ok

4. Inside the Test Class Write Code:
   Step 1 : Write Apex Class
   Step 2 : Click RunTest
   Step 3 : Check Test Class it need to pass
   Step 4 : Check Test Coverage 100%

7. FUNCTIONAL & PERFORMANCE TESTING

Functional Testing

Functional testing ensures that all features of the system work correctly as per the project requirements.

For the project “Replastic Innovations: Transforming Plastic Waste into Sustainable Solutions”, the following functionalities were tested:

Waste Data Entry Testing

Verified that users can successfully enter plastic waste details such as type, quantity, and source without errors.

Recycling Process Tracking

Ensured that the system correctly updates the status of waste from collection to recycling and final product generation.

Validation Rules Testing

Checked that input fields (such as quantity, date, and type of plastic) accept only valid data and display error messages for incorrect entries.

Automation Testing

Tested automated processes such as status updates, notifications, and workflow triggers to ensure smooth operation.

Report Generation Testing

Verified that the system generates accurate reports on waste collection, recycling efficiency, and product output.

Dashboard Testing

Ensured that visual dashboards correctly display real-time data and analytics for decision-making.

User Access Control Testing

Confirmed that different users (admin, staff, manager) have appropriate access permissions.

Performance Testing

Performance testing ensures that the system operates efficiently under different conditions.

Response Time Testing

The system was tested to ensure quick response while entering data and generating reports.

Load Testing

Verified that the system can handle large volumes of waste data and multiple users simultaneously without slowing down.

Scalability Testing

Ensured that the system can support future expansion, such as adding more recycling units or users.

Reliability Testing

Checked that the system consistently performs without crashes or data loss during continuous usage.

Stress Testing

Evaluated system behavior under extreme conditions, such as high data input or peak usage times.

Data Accuracy Testing

Ensured that all calculations and reports maintain high accuracy even with large datasets.

Testing Outcome

- All core functionalities worked as expected.
- The system handled multiple operations efficiently.
- Data accuracy and reliability were maintained.
- Minor issues were identified and resolved during testing.

8. ADVANTAGES & DISADVANTAGES

Advantages

Environmental Protection:

Reduces plastic pollution by converting waste into reusable and eco-friendly products.

Efficient Waste Management:

Provides a systematic approach to collect, segregate, and recycle plastic waste.

Automation and Accuracy:

Minimizes manual errors through automated tracking and data management.

Disadvantages

Initial Setup Cost:

Requires investment in system development, infrastructure, and recycling equipment.

Dependence on Technology:

The system relies on digital platforms, which may face issues like downtime or technical errors.

Internet Dependency:

Continuous internet access is required for real-time updates and system functionality.

9. CONCLUSION

In conclusion, the project “Replastic Innovations: Transforming Plastic Waste into Sustainable Solutions” provides an effective and sustainable approach to managing plastic waste through the use of a structured and automated system. The project successfully integrates waste collection, segregation, recycling, and product generation into a single streamlined process.

By replacing traditional manual methods with a system-driven approach, the project improves efficiency, reduces errors, and ensures accurate tracking of plastic waste. The implementation of automation, reports, and dashboards enables real-time monitoring and better decision-making, which enhances overall operational performance.

Overall, the system demonstrates how technology can be effectively utilized to address real-world environmental challenges. With further enhancements and scalability, Replastic Innovations has the potential to become a large-scale solution for sustainable waste management and eco-friendly development.

10. FUTURE SCOPE

AI-Based Waste Prediction:

Implement artificial intelligence to predict waste generation patterns and optimize recycling processes.

Smart Waste Collection System:

Integration of IoT-enabled smart bins to automatically monitor and collect plastic waste efficiently.

Mobile Application Integration:

Develop a mobile app for users to track waste collection, recycling status, and eco-friendly product availability.

Advanced Recycling Technologies::

Incorporate modern recycling techniques to improve efficiency and produce higher-quality recycled products.

Real-Time Monitoring System:

Enhance dashboards with real-time data analytics for better tracking and decision-making.

11.Appendix

Source code : Salesforce Configuration & Flows

Dataset : Customer and Sales Records

Tools used : Salesforce Developer Org

Conclusion :

The RePlastix Innovations Salesforce CRM project successfully demonstrates how technology can be leveraged to automate and streamline complex business processes in the plastic recycling industry. By integrating custom objects, validation operational challenges such as inventory shortages, delayed restocking, and inefficient order management. The solution not only enhances productivity and data accuracy but also supports real-time decisionmaking through automated alerts and reporting tools. Overall, the project meets its objectives by providing a scalable, user-friendly, and secure CRM tailored to the unique needs of RePlastix Innovations, laying a strong foundation for future enhancements and sustainability-focused growth

Project Resources:

Trailhead Profile:

Demo video:

Github link: